```
lab_9=# -- 1. Store authors
lab_9=# CREATE TABLE authors (
lab_9(#     author_id SERIAL PRIMARY KEY,
lab_9(#     first_name VARCHAR(50) NOT NULL,
lab_9(#     last_name VARCHAR(50) NOT NULL,
lab_9(#     birth_date DATE
lab_9(# );
CREATE TABLE
lab_9=#
lab_9=# -- 2. Store books
lab_9=# CREATE TABLE books (
lab_9(#     book_id SERIAL PRIMARY KEY,
lab_9(#     title VARCHAR(200) NOT NULL,
lab_9(#     isbn VARCHAR(13) UNIQUE,
lab_9(#     publication_year INTEGER,
lab_9(#     available_copies INTEGER DEFAULT 1
lab_9(# );
CREATE TABLE
lab_9=#
lab_9=# -- 3. Connect books and authors (many-to-many)
lab_9=# CREATE TABLE book_authors (
lab_9(#     book_id INTEGER,
lab_9(#     author_id INTEGER,
lab_9(#     PRIMARY KEY (book_id, author_id),
lab_9(#     FOREIGN KEY (book_id) REFERENCES books(book_id),
lab_9(#     FOREIGN KEY (author_id) REFERENCES authors(author_id)
lab_9(# );
CREATE TABLE
```

```
lab_9=# -- 4. Store library members
lab_9=# CREATE TABLE members (
lab_9(#     member_id SERIAL PRIMARY KEY,
lab_9(#     first_name VARCHAR(50) NOT NULL,
lab_9(#     last_name VARCHAR(50) NOT NULL,
lab_9(#     email VARCHAR(100) UNIQUE NOT NULL,
lab_9(#     phone VARCHAR(15),
lab_9(#     membership_date DATE DEFAULT CURRENT_DATE
lab_9(# );
CREATE TABLE
lab_9=#
lab_9=# -- 5. Store borrowing records
lab_9=# CREATE TABLE loans (
lab_9(#     loan_id SERIAL PRIMARY KEY,
lab_9(#     member_id INTEGER NOT NULL,
lab_9(#     book_id INTEGER NOT NULL,
lab_9(#     loan_date DATE DEFAULT CURRENT_DATE,
lab_9(#     due_date DATE NOT NULL,
lab_9(#     return_date DATE,
lab_9(#     late_fee DECIMAL(10,2) DEFAULT 0.00,
lab_9(#     FOREIGN KEY (member_id) REFERENCES members(member_id),
lab_9(#     FOREIGN KEY (book_id) REFERENCES books(book_id)
lab_9(# );
CREATE TABLE
```

```
lab_9=# SELECT table_name
lab_9-# FROM information_schema.tables
lab_9-# WHERE table_schema = 'public'
lab_9-#   AND table_type = 'BASE TABLE'
lab_9-# ORDER BY table_name;
  table_name
--------------
 authors
 book_authors
 books
 loans
 members
(5 rows)


lab_9=# SELECT
lab_9-#     table_name,
lab_9-#     column_name,
lab_9-#     data_type,
lab_9-#     is_nullable
lab_9-# FROM information_schema.columns
lab_9-# WHERE table_schema = 'public'
lab_9-# ORDER BY table_name, ordinal_position;
```

```
  table_name   |    column_name     |      data_type      | is_nullable
---------------+--------------------+---------------------+-------------
 authors       | author_id          | integer             | NO
 authors       | first_name         | character varying   | NO
 authors       | last_name          | character varying   | NO
 authors       | birth_date         | date                | YES
 book_authors  | book_id            | integer             | NO
 book_authors  | author_id          | integer             | NO
 books         | book_id            | integer             | NO
 books         | title              | character varying   | NO
 books         | isbn               | character varying   | YES
 books         | publication_year   | integer             | YES
 books         | available_copies   | integer             | YES
 loans         | loan_id            | integer             | NO
 loans         | member_id          | integer             | NO
 loans         | book_id            | integer             | NO
 loans         | loan_date          | date                | YES
 loans         | due_date           | date                | NO
 loans         | return_date        | date                | YES
 loans         | late_fee           | numeric             | YES
 members       | member_id          | integer             | NO
 members       | first_name         | character varying   | NO
 members       | last_name          | character varying   | NO
 members       | email              | character varying   | NO
 members       | phone              | character varying   | YES
 members       | membership_date    | date                | YES
(24 rows)
```

```
lab_9=# INSERT INTO books (title, isbn, publication_year, available_copies)
lab_9-# VALUES
lab_9-#     ('1984', '9780451524935', 1949, 4),
lab_9-#     ('Animal Farm', '9780451526342', 1945, 3),
lab_9-#     ('Harry Potter and the Philosopher''s Stone', '9780747532699', 1997, 5),
lab_9-#     ('The Hobbit', '9780547928227', 1937, 2),
lab_9-#     ('The Lord of the Rings', '9780544003415', 1954, 3),
lab_9-#     ('Murder on the Orient Express', '9780062693662', 1934, 2),
lab_9-#     ('War and Peace', '9780199232765', 1869, 4),
lab_9-#     ('Pride and Prejudice', '9780141439518', 1813, 3);
INSERT 0 8
lab_9=# SELECT * FROM books;
 book_id |                 title                  |     isbn      | publication_year | available_copies
---------+----------------------------------------+---------------+------------------+------------------
       1 | Pride and Prejudice                    |               |             1813 |                3
       2 | 1984                                   | 9780451524935 |             1949 |                4
       3 | Animal Farm                            | 9780451526342 |             1945 |                3
       4 | Harry Potter and the Philosopher's Stone | 9780747532699 |             1997 |                5
       5 | The Hobbit                             | 9780547928227 |             1937 |                2
       6 | The Lord of the Rings                  | 9780544003415 |             1954 |                3
       7 | Murder on the Orient Express           | 9780062693662 |             1934 |                2
       8 | War and Peace                          | 9780199232765 |             1869 |                4
       9 | Pride and Prejudice                    | 9780141439518 |             1813 |                3
(9 rows)
```

```
lab_9=# INSERT INTO members (first_name, last_name, email, phone)
lab_9-# VALUES
lab_9-#     ('Emma', 'Johnson', 'emma.johnson@example.com', '555-0101'),
lab_9-#     ('Michael', 'Brown', 'michael.brown@example.com', '555-0102'),
lab_9-#     ('Sophia', 'Davis', 'sophia.davis@example.com', '555-0103'),
lab_9-#     ('Daniel', 'Wilson', 'daniel.wilson@example.com', '555-0104'),
lab_9-#     ('Olivia', 'Taylor', 'olivia.taylor@example.com', '555-0105');
INSERT 0 5
lab_9=# SELECT * FROM members;
 member_id | first_name | last_name |          email          |  phone   | membership_date
-----------+------------+-----------+-------------------------+----------+-----------------
         1 | Alex       | Smith     | alex.test@example.com   |          | 2026-10-12
         2 | Emma       | Johnson   | emma.johnson@example.com | 555-0101 | 2026-10-12
         3 | Michael    | Brown     | michael.brown@example.com | 555-0102 | 2026-10-12
         4 | Sophia     | Davis     | sophia.davis@example.com | 555-0103 | 2026-10-12
         5 | Daniel     | Wilson    | daniel.wilson@example.com | 555-0104 | 2026-10-12
         6 | Olivia     | Taylor    | olivia.taylor@example.com | 555-0105 | 2026-10-12
(6 rows)
```

```
lab_9=# SELECT 'authors' AS table_name, COUNT(*) AS total
lab_9-# FROM authors
lab_9-#
lab_9-# UNION ALL
lab_9-#
lab_9-# SELECT 'books', COUNT(*)
lab_9-# FROM books
lab_9-#
lab_9-# UNION ALL
lab_9-#
lab_9-# SELECT 'members', COUNT(*)
lab_9-# FROM members;
 table_name | total
------------+-------
 authors    |     5
 books      |     9
 members    |     6
(3 rows)
```